

39 Plant Responses to Internal and External Signals

KEY CONCEPTS

- 39.1** Signal transduction pathways link signal reception to response *p. 843*
- 39.2** Plants use chemicals to communicate *p. 845*
- 39.3** Responses to light are critical for plant success *p. 855*
- 39.4** Plants respond to a wide variety of stimuli other than light *p. 861*
- 39.5** Plants respond to attacks by pathogens and herbivores *p. 866*

Study Tip

Make a table: As you read the chapter, add specific examples for each of the general categories of responses shown in the diagram.

Factor	Example of plant response
Light	Seed germination in response to red light

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For Students (in eText and Study Area)

- Get Ready for Chapter 39
- Video: Gravitropism
- Video: *Mimosa* leaves

For Instructors to Assign (in Item Library)

- Activity: Leaf Abscission
- Activity: Plant Hormones



Figure 39.1 Sunflowers track the sun from east to west each day. After sunset, they reverse direction, facing the direction of the next sunrise. By facing the hot sun during the day, the floral heads become warmer and release greater amounts of chemicals that attract pollinators. Light is just one of the many factors to which a plant responds.

